STRATIFIED RISK PROFILING AND COGNITIVE DECLINE PATTERNS IN ALZHEIMER'S DISEASE USING NACC-UDS DATA

Vaishnavi Tiwari

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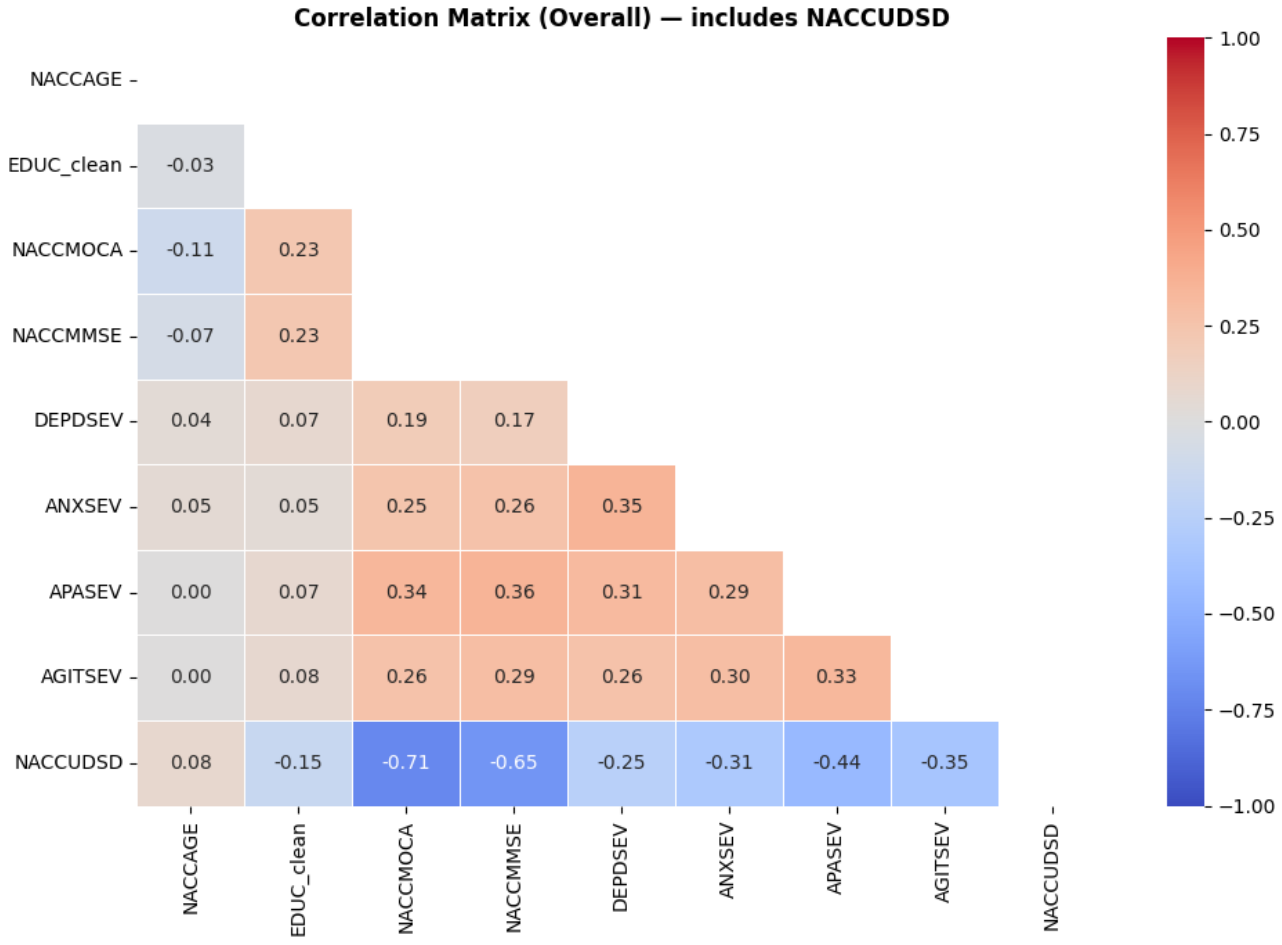
Data Source: NACC Uniform Data Set (UDS) | investigator_nacc72.csv
207,454 visit records | 55,268 patients | 1,024 variables | 2005 to 2025

Abstract

Alzheimer's Disease (AD) is the leading cause of dementia worldwide, imposing an estimated 360 billion dollars in annual care expenditure in the United States alone. Despite substantial clinical data collection, AD progression remains highly heterogeneous. This project applies stratified exploratory data analysis, supervised machine learning, and SHAP-based model interpretability to the NACC Uniform Data Set (UDS), encompassing 207,454 visit-level records from 55,268 unique participants from June 2005 to October 2025.

The goal is three-class cognitive status classification (Normal, MCI, Dementia) using 13 clinical features including MMSE, MoCA, longitudinal trajectory slope features, behavioral neuropsychiatric symptoms, demographics, and the APOE epsilon 4 genetic risk marker. SMOTENC addressed MCI class imbalance at 27% of baseline visits.

Ten models were trained and evaluated. The best 3-class model, Tuned Gradient Boosting with SHAP-guided interaction features, achieved 72.5% test accuracy compared to a 41.4% majority-class baseline, a gain of 31 percentage points. SHAP confirmed MMSE and MoCA as dominant predictors while apathy severity emerged as a strong third signal. MCI classification achieves up to 59% recall (Neural Network), reflecting genuine clinical ambiguity at the MCI boundary.



1. Introduction

1.1 Background and Motivation

Alzheimer's Disease affects an estimated 6.9 million Americans aged 65 and older, with global prevalence expected to reach 139 million by 2050. Unlike many chronic conditions, AD has no disease-modifying treatment, making early accurate identification of at-risk individuals essential for clinical trial recruitment, care planning, and therapeutic intervention.

Standard clinical assessment relies primarily on MMSE and MoCA cognitive test scores supplemented by clinician judgment. These instruments do not capture the longitudinal shape of cognitive decline, which may contain substantial additional prognostic signal. This project addresses three gaps: (1) existing analyses tend to ignore trajectory and use single-visit data, (2) class imbalance between Normal, MCI, and Dementia is frequently unaddressed, and (3) black-box model outputs are not interpretable in clinical contexts.

1.2 Research Questions

- RQ1: Can machine learning models classify patients into Normal, MCI, and Dementia accurately from routine clinical data?
- RQ2: Which features are most predictive, and do SHAP importances align with EDA correlations?
- RQ3: Does SMOTENC meaningfully improve MCI recall?
- RQ4: Do stacking ensembles outperform individual models?
- RQ5: Is there a cognitive reserve effect where education moderates cognitive score and disease status?

1.3 Dataset Overview

Category	Visit Count	Visit %	Person Count	Person %
1 Normal	101,837	49.1%	22,863	41.4%
2 Impaired (not MCI)	9,184	4.4%	2,496	4.5%
3 MCI	36,637	17.7%	12,466	22.6%
4 Dementia	59,796	28.8%	17,443	31.6%
Total	207,454	100%	55,268	100%

2. Literature Review

2.1 Longitudinal Assessment of Cognitive Decline in High-Level Alzheimer Disease Neuropathologic Change

Source: NACC longitudinal cohort. PMC Article PMC12676791.

This study examined 586 adults with confirmed high-level AD neuropathologic change grouped into three trajectory profiles. Key findings: rapid decline was associated with APOE epsilon 4, higher tau and amyloid burden, and white matter damage. Cognitive resilience was associated with lower genetic risk and better systemic health management. This paper validates the inclusion of NACCNE4S_first (APOE marker) and motivates the MOCA_slope and MMSE_slope trajectory features in this project's feature set.

2.2 Machine Learning for AD Progression Prediction on ADNI Data

Source: Falahati et al. (2014). Journal of Alzheimer's Disease, 41(4), 1071-1083.

Model	Accuracy	MCI Recall
Random Forest	89%	61%
Gradient Boosting	87%	61%
SVM	84%	approximately 58%
Logistic Regression	80%	approximately 50%

Top features were MMSE, ADAS-Cog, APOE, and age. MCI recall at 61% was the lowest class. Class imbalance was not addressed. This project extends Falahati et al. by applying SMOTENC, adding longitudinal slope features, using SHAP interpretability, and operating on a dataset 85 times larger.

3. Exploratory Data Analysis

3.1 Target Variable Distribution

Code: NACCUDSD Distribution

```
NACC_MISSING = [-1,-2,-3,-4,-6,-7,-8,-9,-99,-999]
df[clean_cols] = df[clean_cols].replace(NACC_MISSING, np.nan)
df["NACCMOCA"] = df["NACCMOCA"].replace([88,99], np.nan)
df["NACCMMSE"] = df["NACCMMSE"].replace([88,95,96,97,98], np.nan)
for col in ["NACCAPOE","NACCNE4S"]: df[col] = df[col].replace(9, np.nan)
baseline = df.sort_values(["NACCID","VISIT_ORDER"]).groupby("NACCID").first()
print(df["NACCUDSD"].value_counts().sort_index())
```

Output:

```
Visit-level:  1=101,837  2=9,184  3=36,637  4=59,796
Person-level: 1=22,863   2=2,496  3=12,466  4=17,443
```

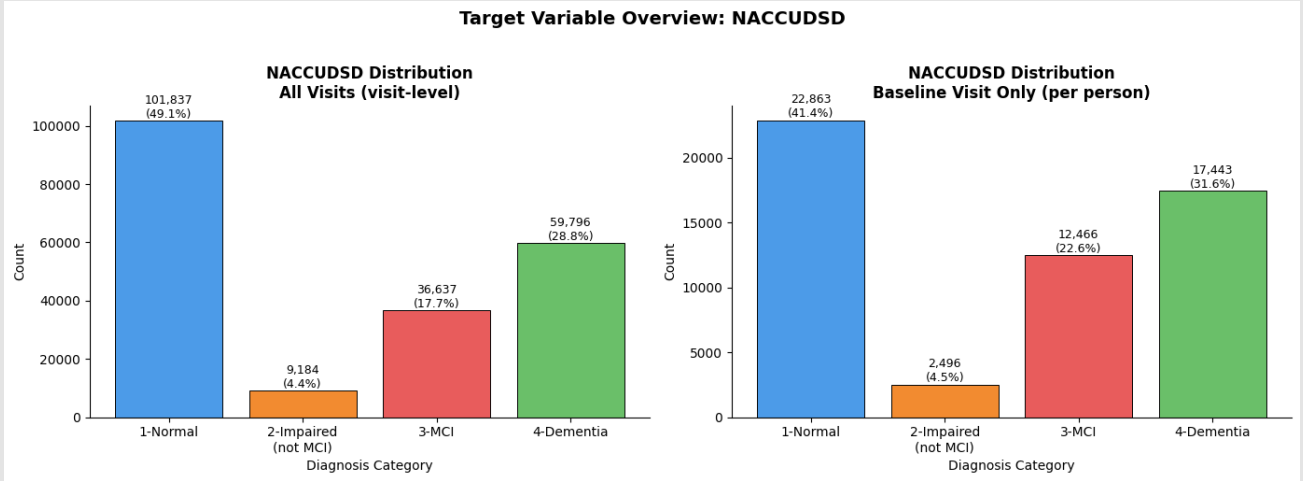


Figure 1: NACCUDSD distribution. Left: all 207,454 visit records. Right: baseline visit per patient (55,268 total). The Normal class dominates at 41 to 49% depending on view. MCI is a minority class at 23 to 27%.

3.2 Missingness Analysis

Output: Missingness Rates

```
NACCMOCA     63.9%  (rises to 77% in Dementia group  non-random, clinically expected)
NACCMMSE     52.1%  (more uniform across groups)
NACCNE4S     13.7%  (74% coverage  workable)
DEPDSEV       6.5%  |  ANXSEV 6.4%  |  APASEV 6.4%  |  AGITSEV 6.3%
EDUC_clean    0.5%  (near complete)
```

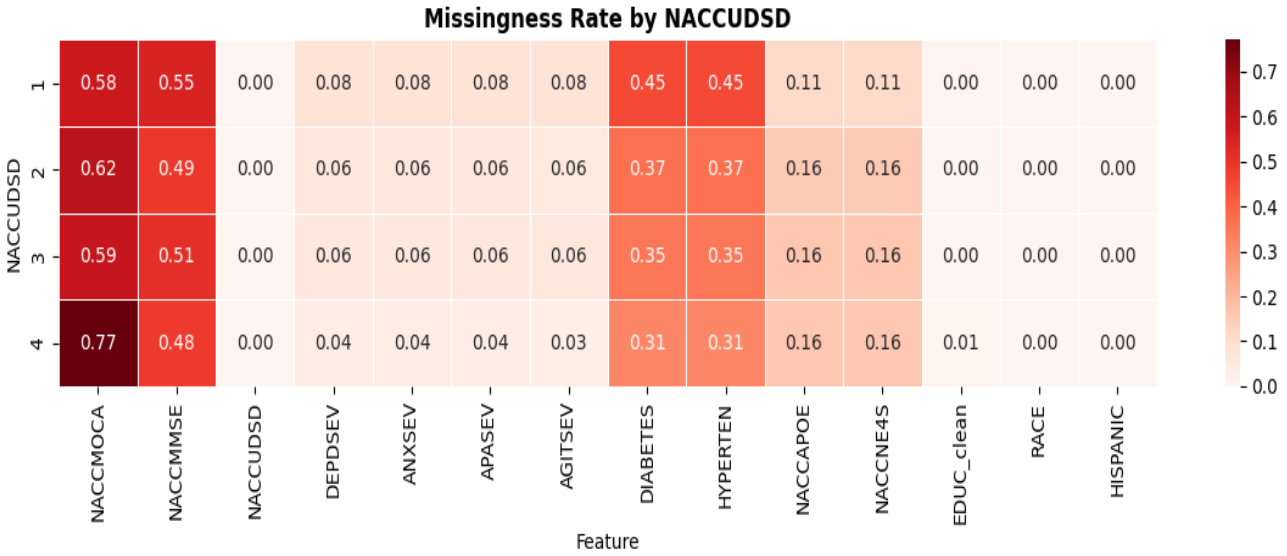


Figure 2: Missingness heatmap by NACCUDSD group. Darker red indicates higher missingness. MoCA missing 77% in Dementia patients because severely impaired patients cannot complete the 30-item test. Behavioral symptoms are the most complete features at 4 to 8% missing.

3.3 Feature Distributions by Disease Group

Output: Mean Values by NACCUDSD

```
           NACCAGE  EDUC  NACCMOCA  NACCMMSE  DEPDSEV  ANXSEV  APASEV  AGITSEV
Normal      73.46  16.05    26.50     28.91     7.15    7.36    7.71     7.65
Impaired    73.70  15.28    24.40     28.00     6.30    6.66    7.06     7.04
MCI         75.52  15.50    22.52     27.03     6.09    6.41    6.81     6.87
Dementia    75.13  14.91    14.99     19.18     5.56    5.36    4.98     5.59

MoCA cliff MCI to Dementia: 22.52 -> 14.99  (7.5 point drop)
MMSE cliff MCI to Dementia: 27.03 -> 19.18  (7.9 point drop)
```

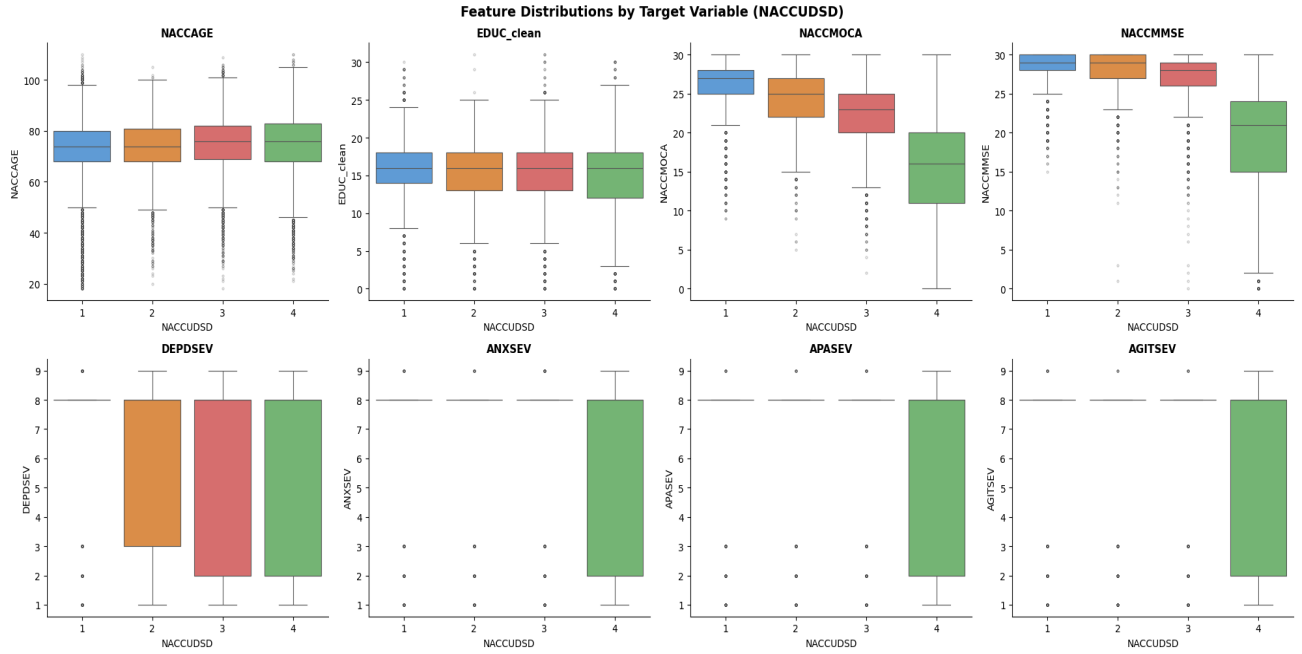


Figure 3: 2x4 boxplot grid showing feature distributions by NACCUDSD group. MoCA and MMSE show the sharpest group separation with a dramatic cliff between MCI and Dementia. Age shows minimal separation. Behavioral symptoms show consistent directional patterns with wide within-group variance.

3.4 Cognitive Score Distributions

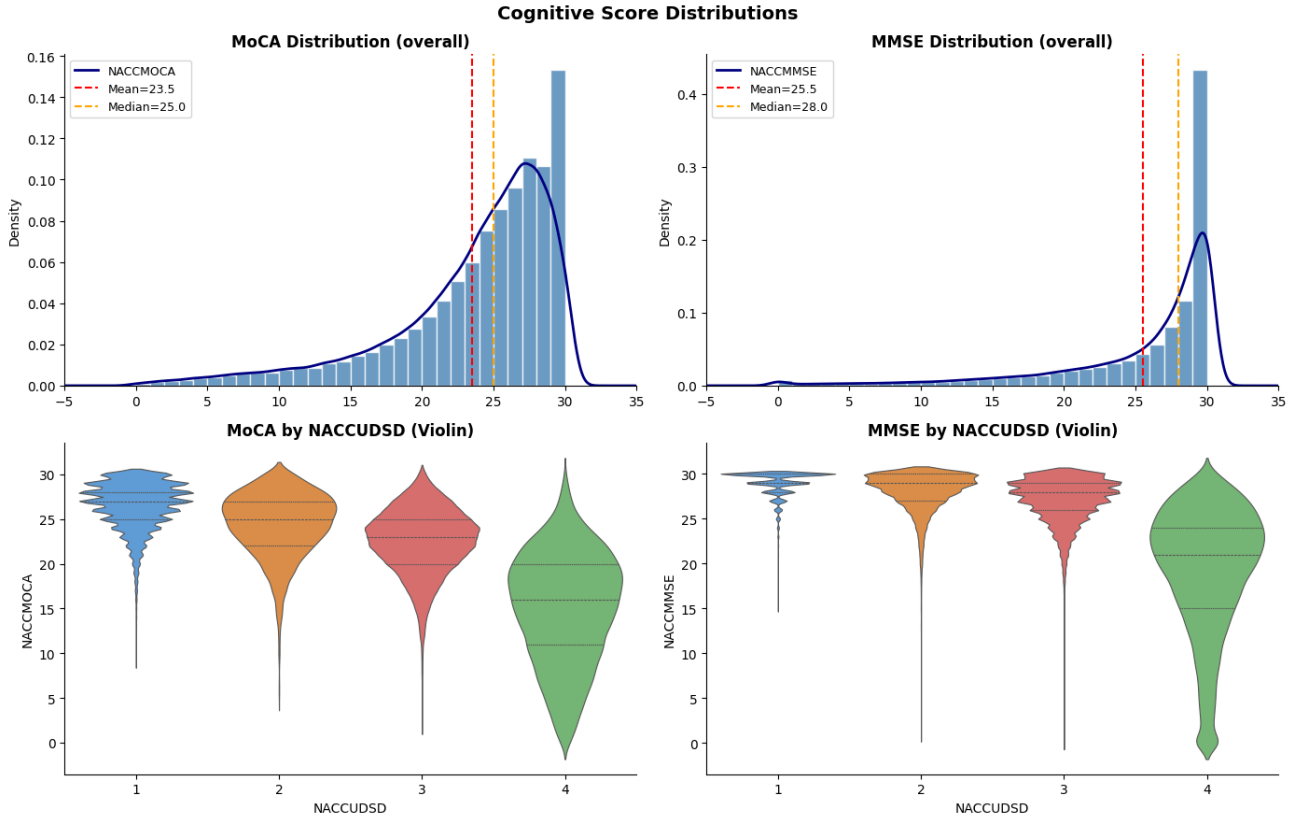


Figure 4: MoCA and MMSE distributions. Top row: overall histograms with KDE overlay. Both are left-skewed; Dementia patients pull the mean below the median. Bottom row: violin plots by NACCUDSD group. Groups 1 to 3 are narrow and top-heavy. Group 4 (Dementia) has a wide violin extending toward zero, reflecting massive within-group heterogeneity.

3.5 Demographic Breakdown by Disease Group

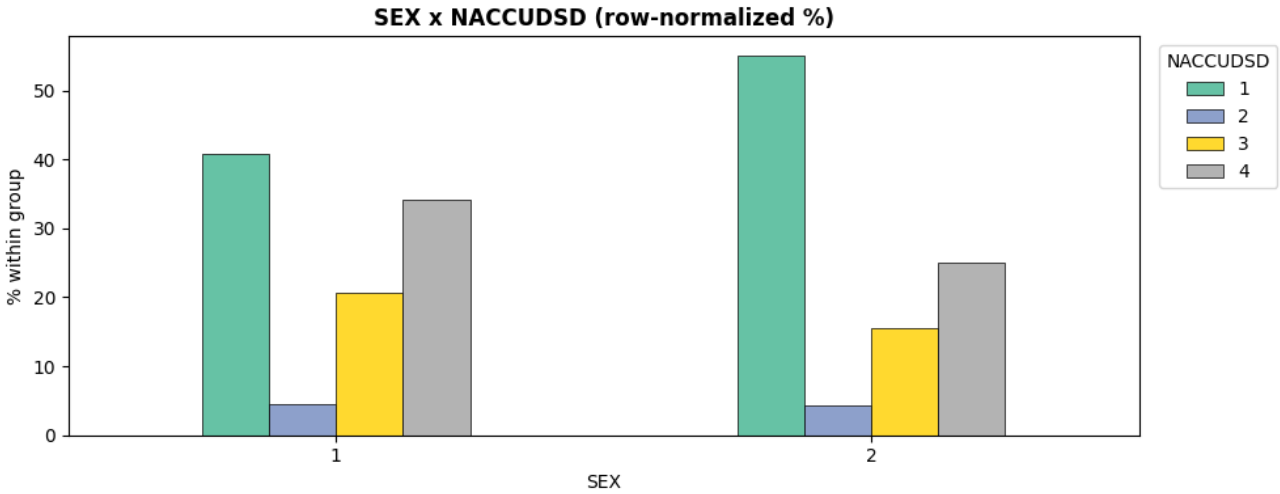


Figure 5: SEX distribution by NACCUDSD. Male vs. Female proportions across cognitive status groups.

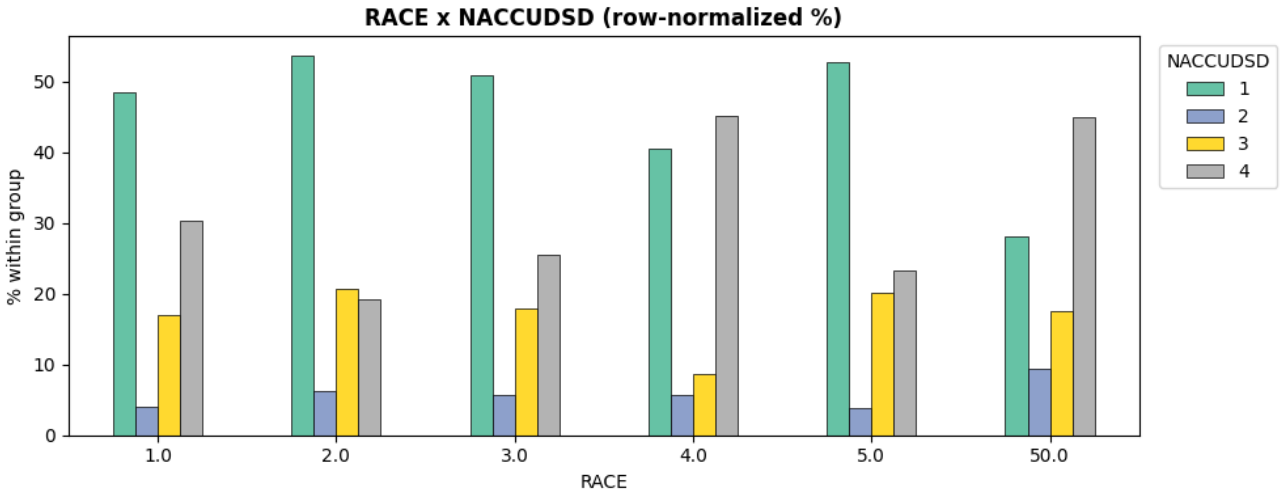


Figure 6: RACE distribution by NACCUDSD. Row-normalized proportions of each racial category across cognitive status groups.

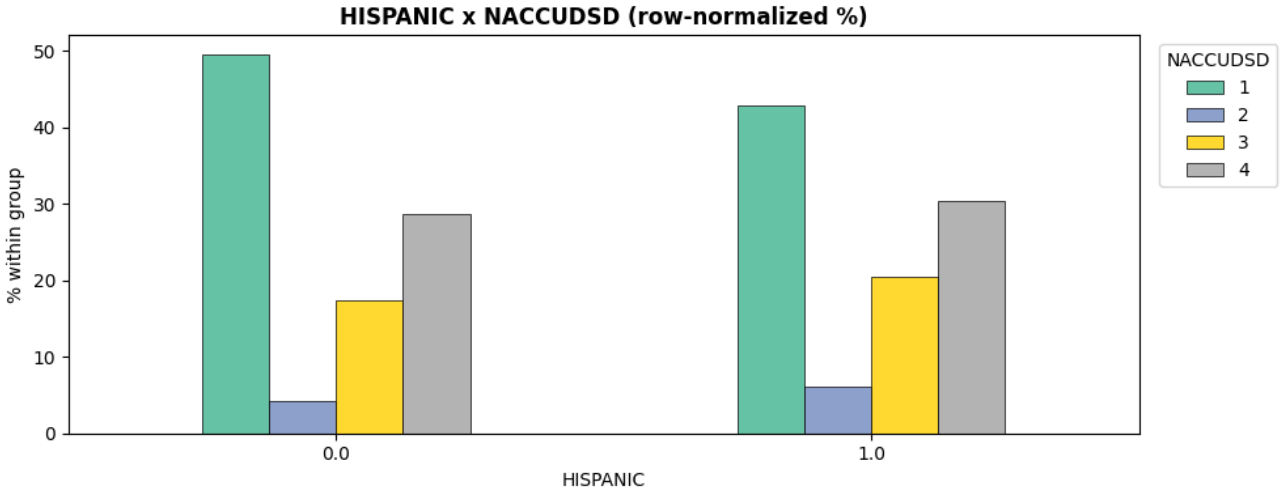


Figure 7: HISPANIC ethnicity distribution by NACCUDSD. Nearly identical distributions between groups confirming ethnicity is not a meaningful differentiator in this cohort.

3.6 Correlation Analysis

Output: Pearson Correlations with NACCUDSD

```
NACCMOCA    -0.71  (strongest predictor)
NACCMMSE    -0.65
APASEV      -0.44  (unexpectedly strong, rank 3)
DEPDSEV     -0.35  |  ANXSEV -0.32  |  AGITSEV -0.30
EDUC_clean  -0.15  (cognitive reserve signal)
NACCAGE      0.08  (near-zero, weakest predictor)
```

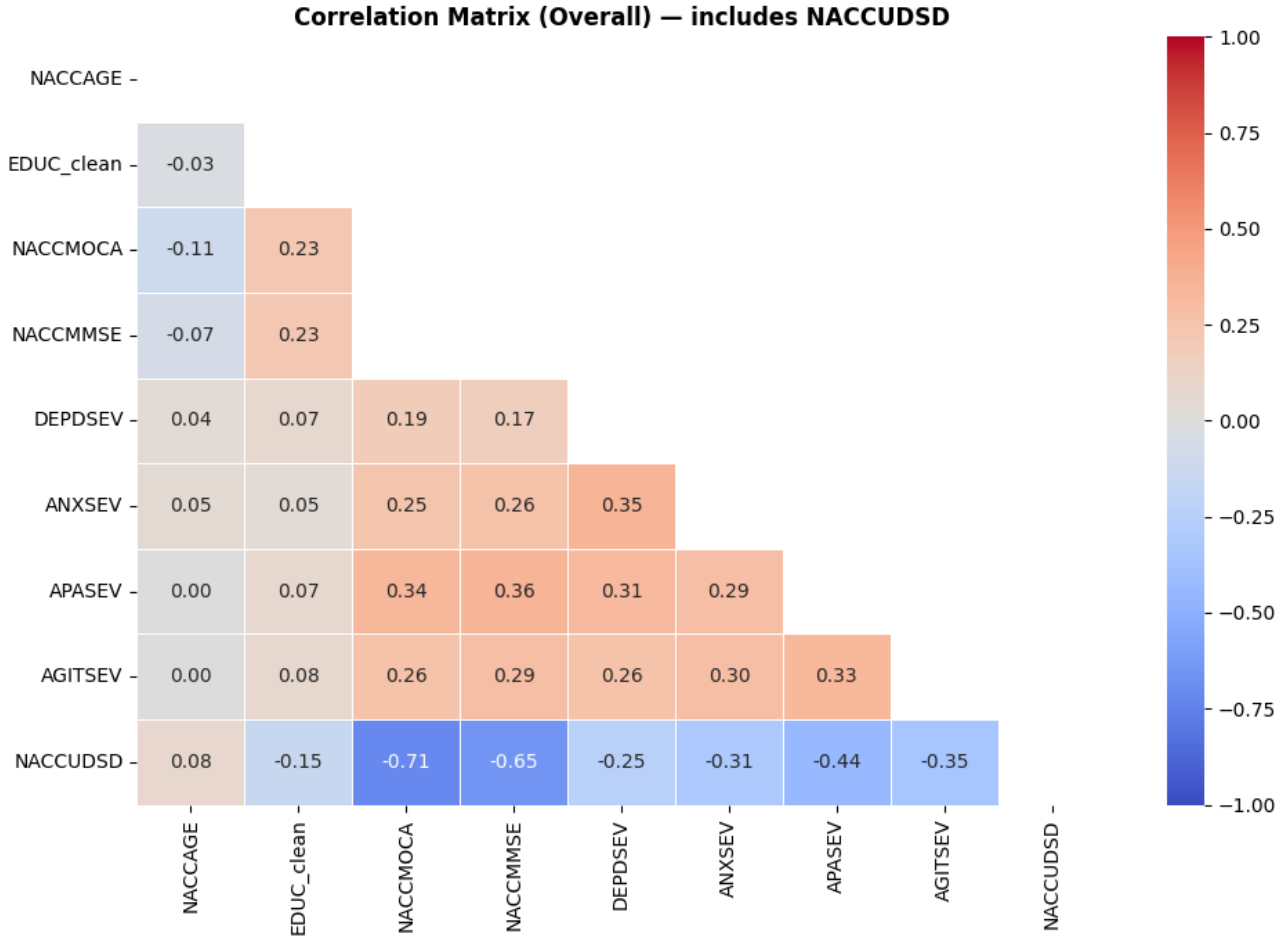


Figure 8: Overall correlation matrix. MoCA and MMSE dominate. Behavioral symptoms show consistent moderate correlations with NACCUDSD.

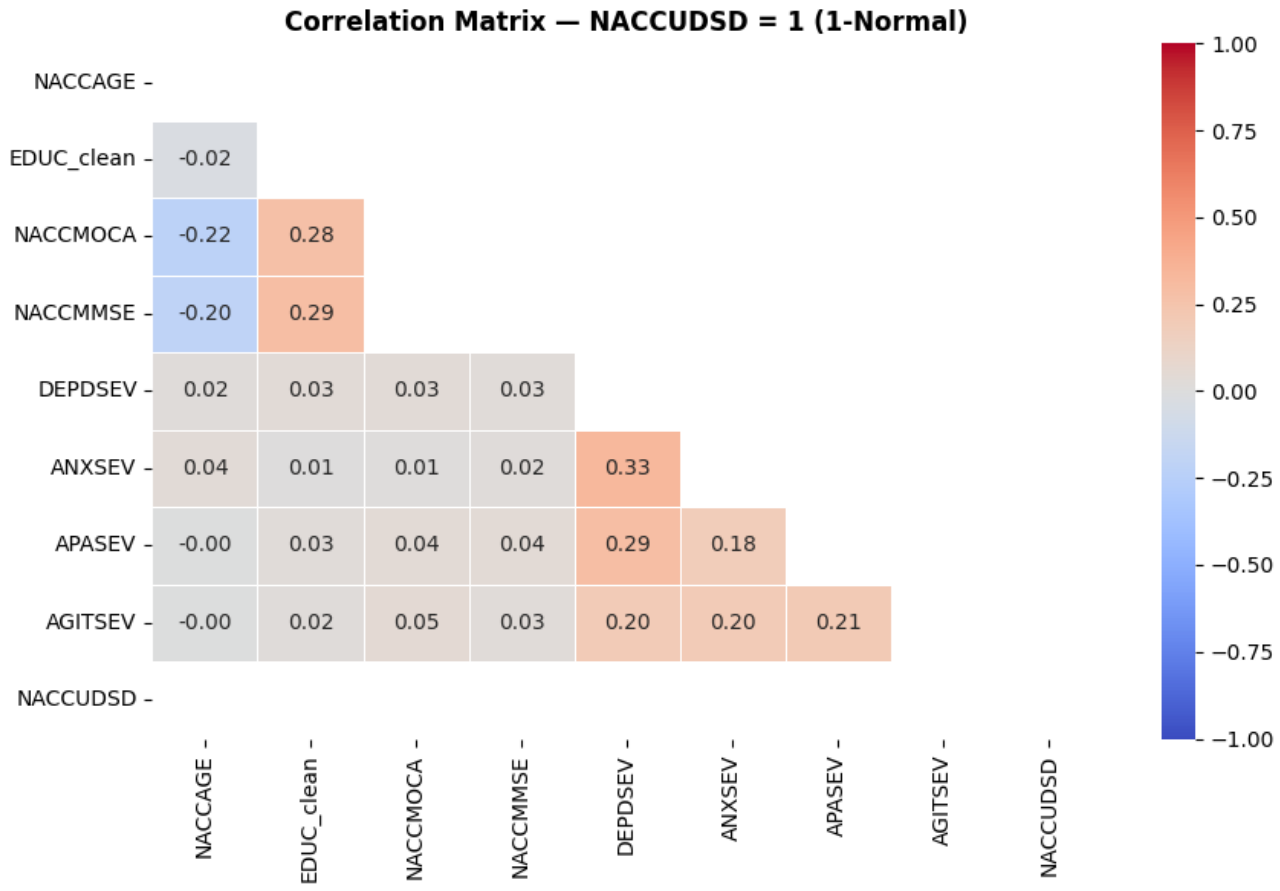


Figure 9: Correlation matrix stratified for Group 1 (Normal). Education vs. MoCA correlation = 0.28.

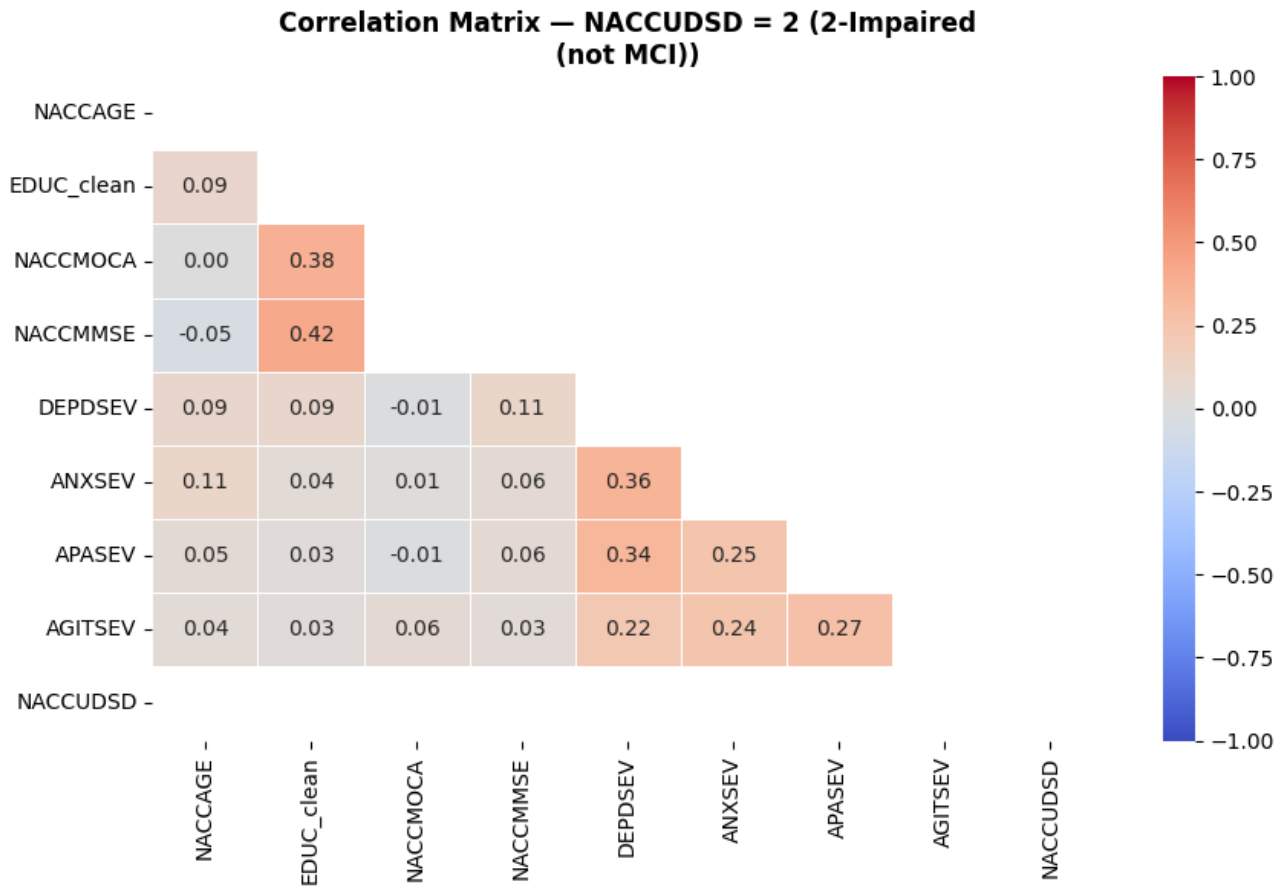


Figure 10: Correlation matrix for Group 2 (Impaired not MCI). Education vs. MoCA peaks at 0.38, the highest of any group. This is the cognitive reserve effect: education buffers most during the transitional stage, answering RQ5.

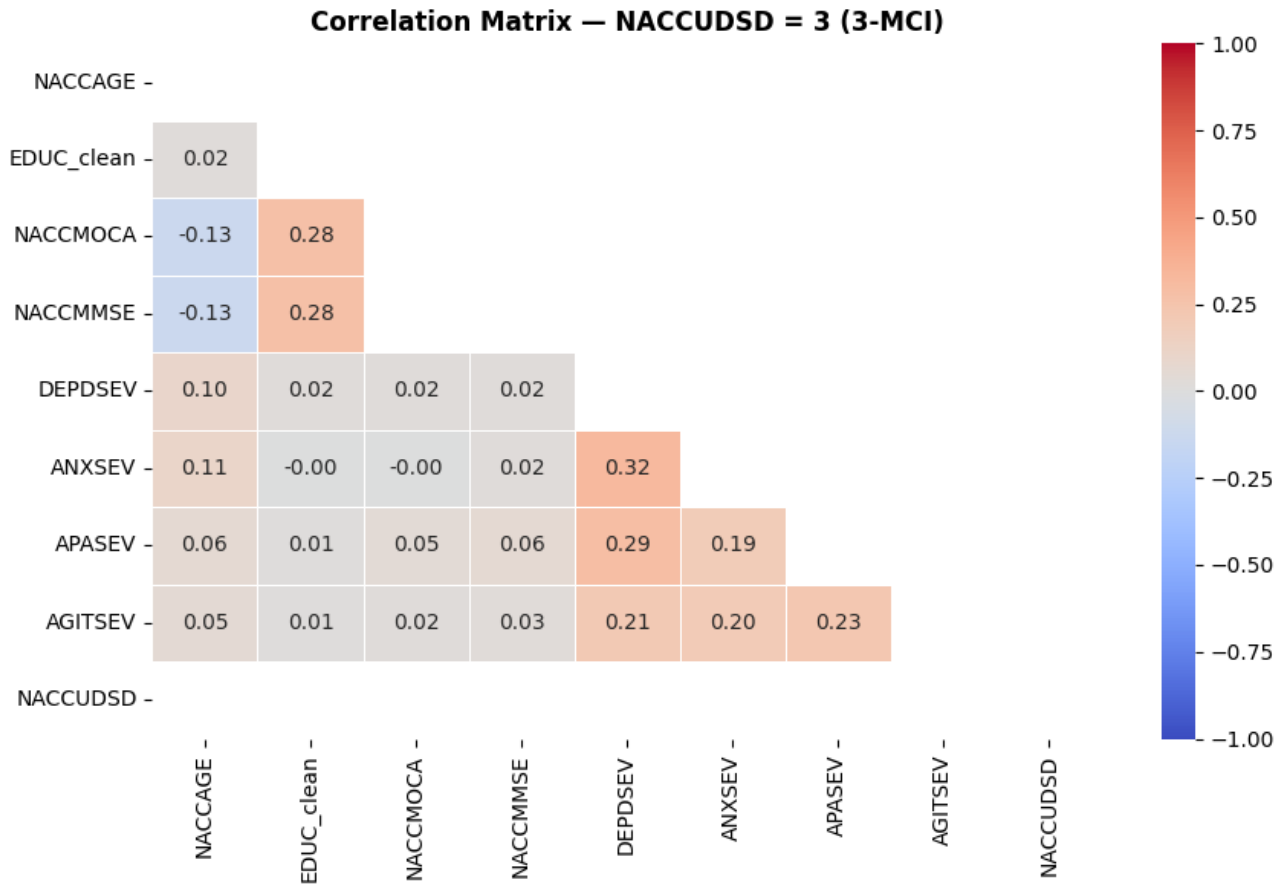


Figure 11: Correlation matrix for Group 3 (MCI). Education vs. MoCA returns to 0.28.

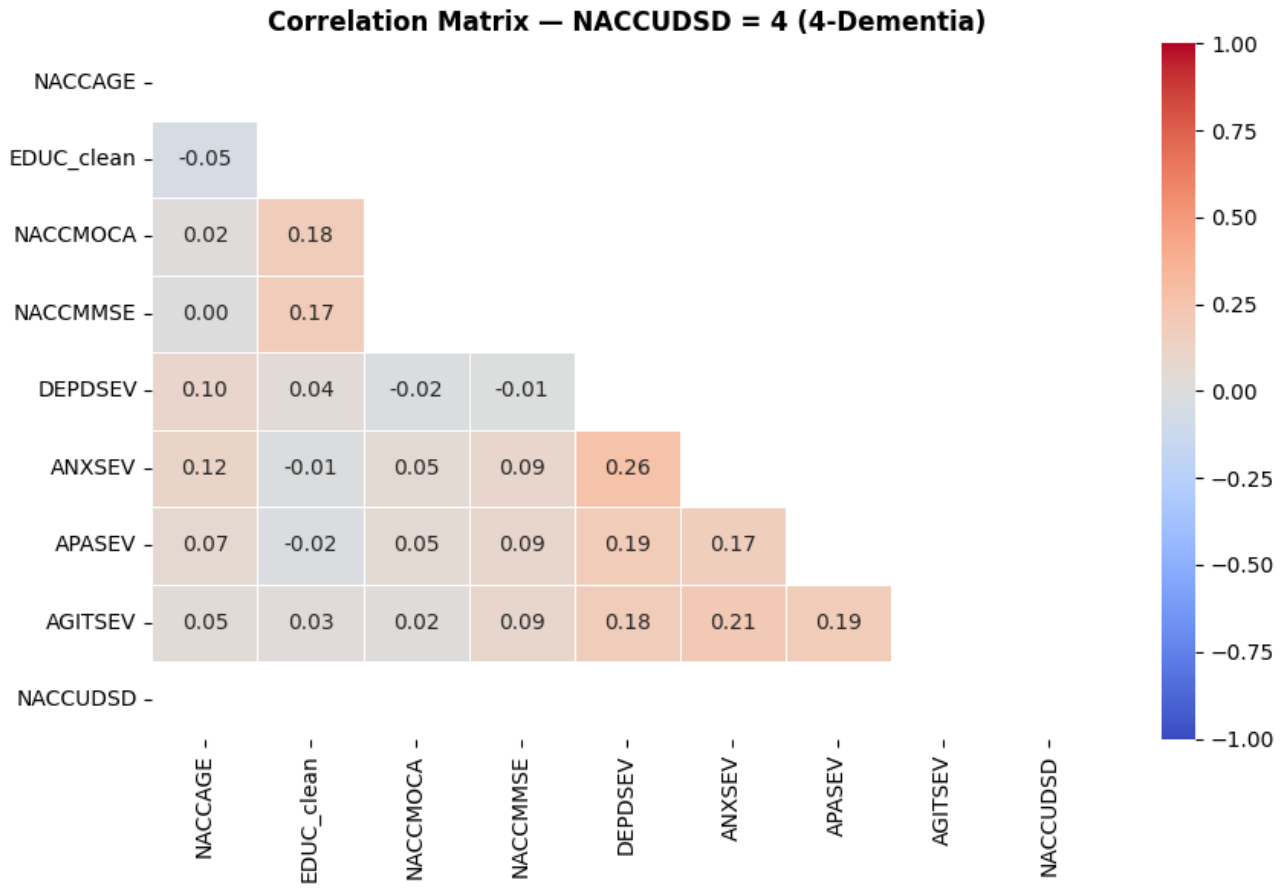


Figure 12: Correlation matrix for Group 4 (Dementia). Education vs. MoCA weakens to 0.18. By full dementia, education can no longer compensate. Partially confirms RQ5 (cognitive reserve effect).

3.7 Age and Education Scatter Plots

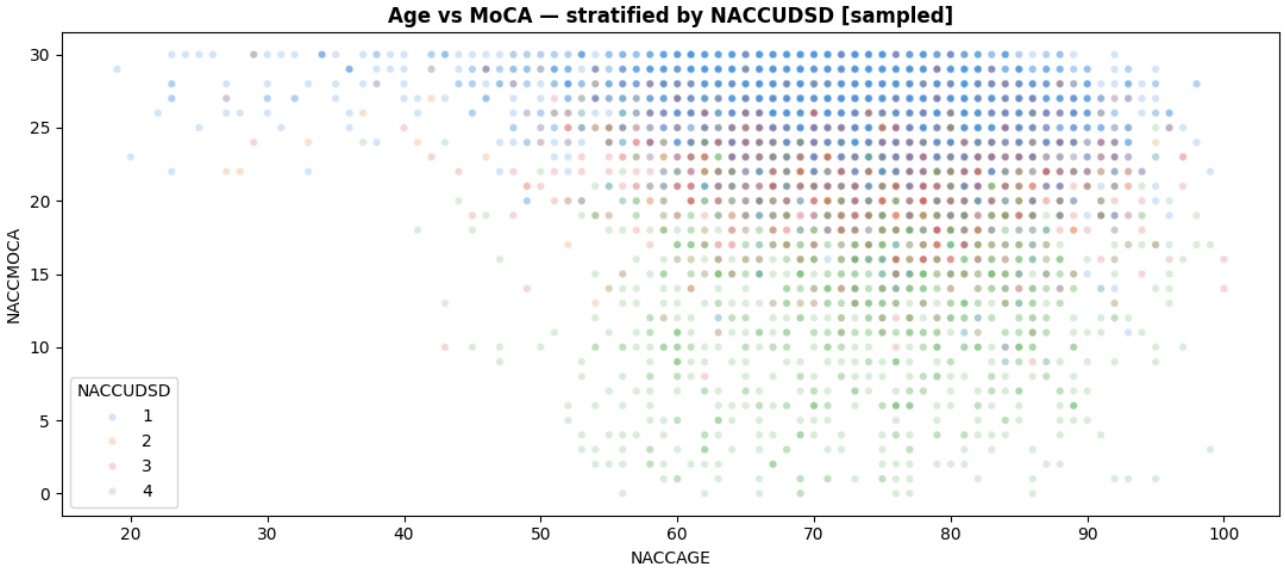


Figure 13: Age vs. MoCA stratified by NACCUDSD. Age is near-zero correlated with MoCA overall. Within the Dementia group, the scatter is wide and flat confirming that age alone does not predict cognitive status.

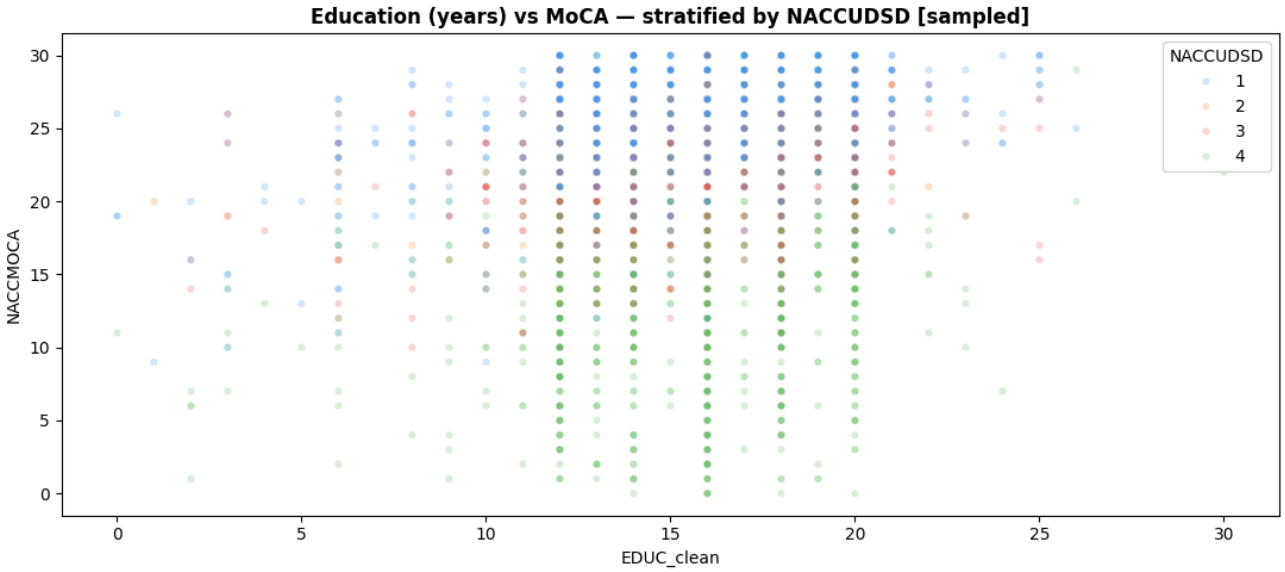


Figure 14: Education vs. MoCA stratified by NACCUDSD. A positive trend is visible especially in Normal and MCI groups. The Dementia group shows a compressed distribution at lower MoCA values regardless of education level.

3.8 Transition Matrix

Output: Adjacent Visit Transition Matrix (row-normalized)

```
                    -> Normal  -> Impaired  -> MCI  -> Dementia
Normal (1)              0.915     0.024     0.054     0.007   (92% stay)
Impaired (2)            0.241     0.504     0.200     0.055   (50% stay  most unstable)
MCI (3)                 0.101     0.043     0.675     0.181   (18% convert to
Dementia/visit)
Dementia (4)            0.004     0.005     0.018     0.973   (97% stay  one-way door)
```

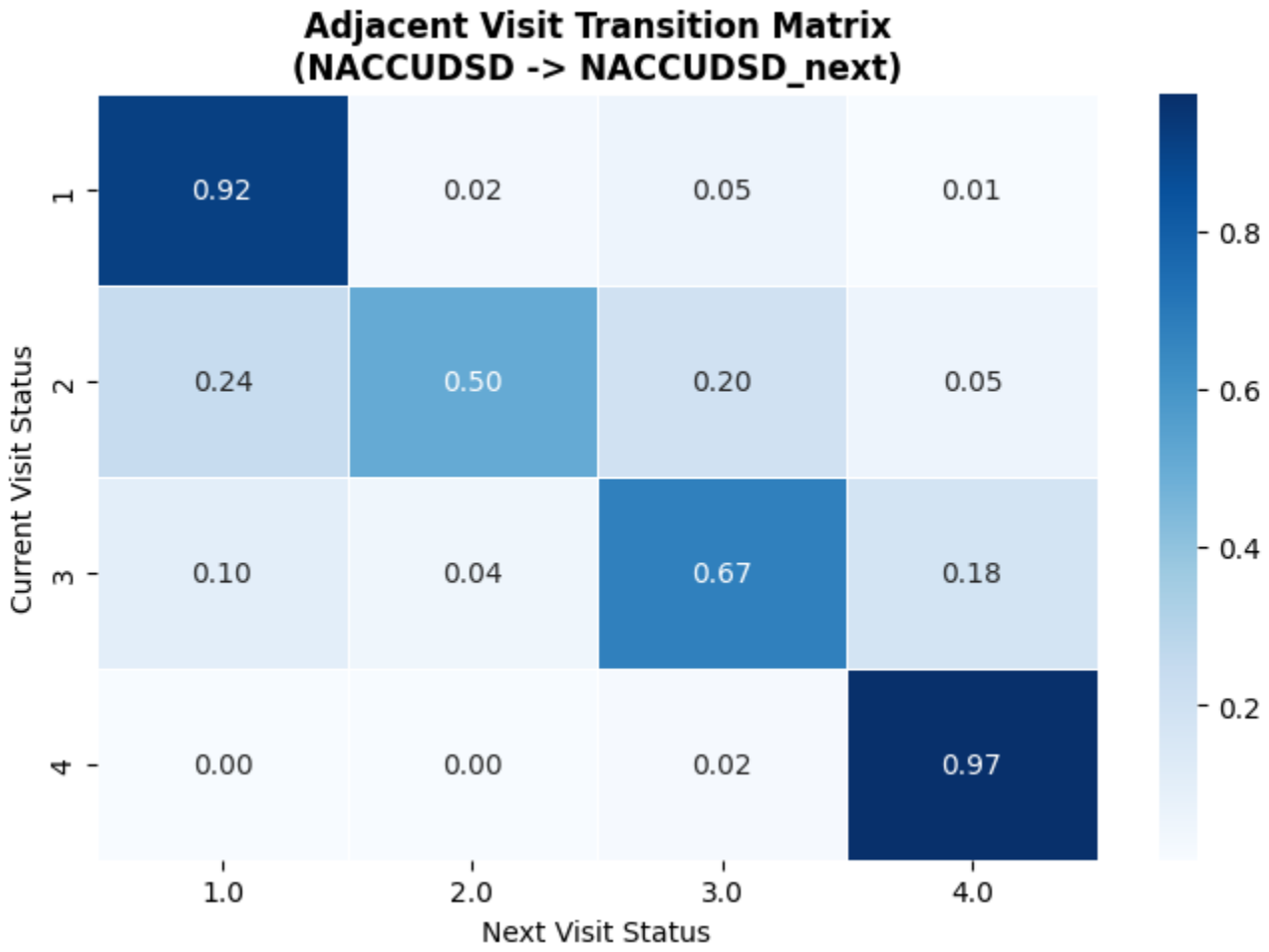


Figure 15: Adjacent visit transition heatmap. The diagonal dominates showing most patients stay in their current state. Dementia is functionally irreversible at 97% stay rate. Group 2 has the lightest diagonal at 50%, confirming it should be merged into MCI for modeling.

3.9 Longitudinal Cognitive Change

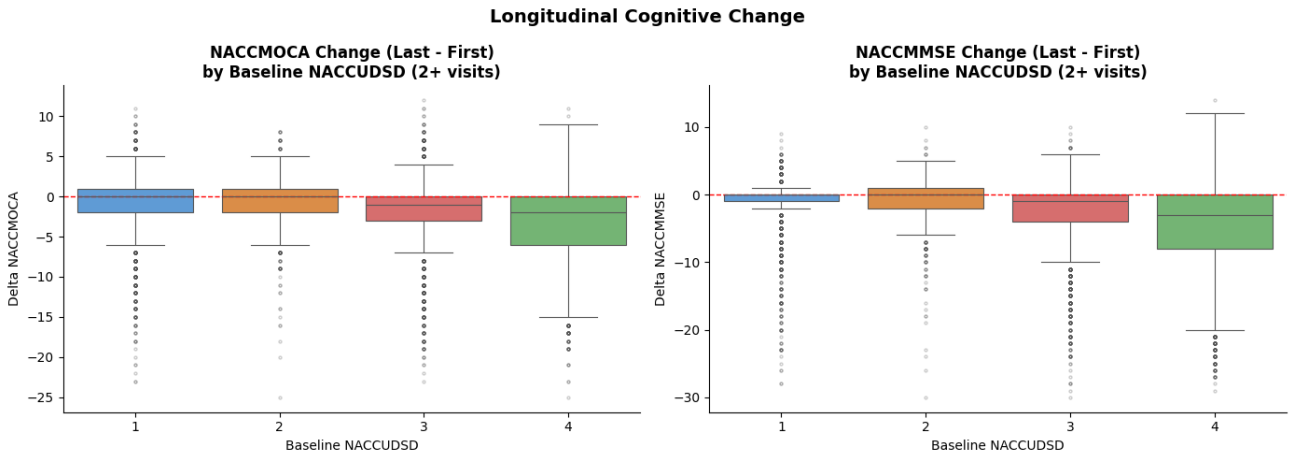
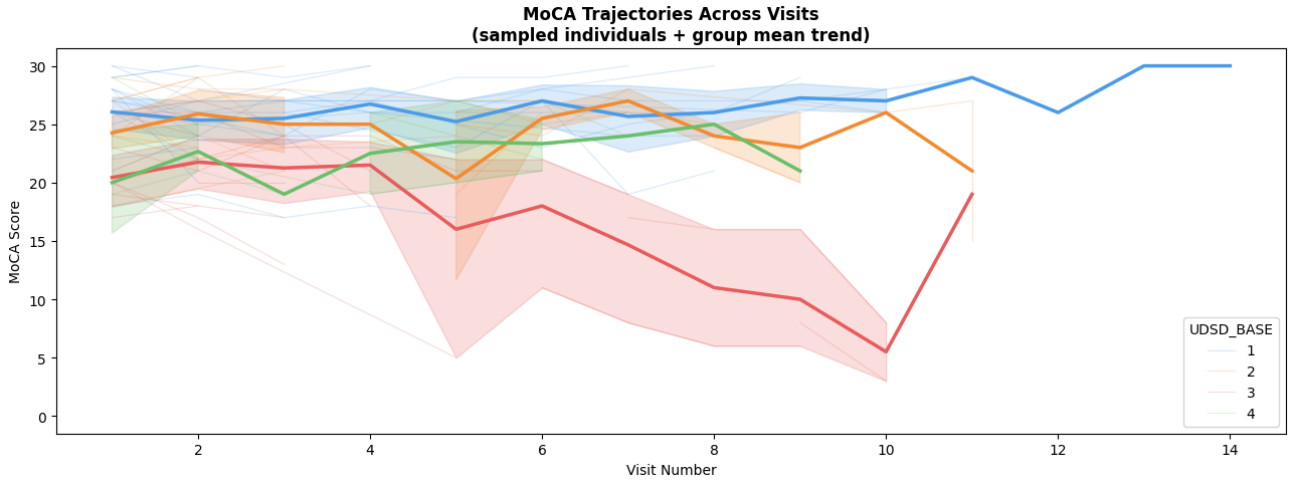


Figure 16: MoCA and MMSE change (last minus first visit score) by baseline NACCUDSD group. Red dashed line at zero separates improvement from decline. Normal patients cluster near zero. Dementia patients show steep negative deltas, with whiskers extending to -20 on MoCA and -30 on MMSE for patients who started moderate and declined to near zero across multiple visits.

3.10 Sample Cognitive Trajectories

Figure 17: Individual MoCA trajectories over visits (40 sampled patients per group). Normal (blue): flat line near 27 to 30 across all visits. MCI (green): holds steady through early visits then drifts downward, defining a window for behavioral intervention. Dementia (red): steep continuous decline to near zero by visit 10 to 11. The Dementia confidence band widens dramatically, reflecting high within-group heterogeneity in decline rate. This trajectory shape difference directly motivates MOCA_slope and MMSE_slope as engineered features.



4. Feature Engineering and Preprocessing

4.1 Class Merging and Target Definition

Group 2 (Impaired not MCI) constitutes only 4% of patients and shows a 50% stay rate in the transition matrix, too unstable to model separately. It was merged into Group 3 (MCI). First-visit diagnosis was used as the prediction target to represent baseline cognitive status.

Code: Merge Group 2 and Define TARGET

```
model_df["NACCUDSD_first"] = model_df["NACCUDSD_first"].replace(2, 3)
model_df["TARGET"] = model_df["NACCUDSD_first"].astype(int)
```

Output:

```
TARGET 1 (Normal):   22,863  (41.4%)
TARGET 3 (MCI):      14,962  (27.1%)  includes merged Group 2
TARGET 4 (Dementia): 17,443  (31.6%)
```

4.2 Cognitive Slope Features

Code: MOCA_slope and MMSE_slope

```
model_df["MOCA_slope"] = (
    (model_df["NACCMOCA_last"] - model_df["NACCMOCA_first"]) /
    model_df["n_visits"].clip(lower=1))
model_df["MMSE_slope"] = (
    (model_df["NACCMMSE_last"] - model_df["NACCMMSE_first"]) /
    model_df["n_visits"].clip(lower=1))
```

Output:

```
MOCA_slope mean: -0.19 points per visit
MMSE_slope mean: -0.42 points per visit
```

4.3 Final Feature Set: 13 Features

Cognitive: NACCMOCA_first, NACCMMSE_first, MOCA_slope, MMSE_slope. Demographics: NACCAGE_first, EDUC_clean_first, SEX_first. Genetic: NACCNE4S_first. Utilization: n_visits. Behavioral: DEPDSEV_first, ANXSEV_first, APASEV_first, AGITSEV_first.

4.4 Train/Test Split, Imputation, and Scaling

Code: Stratified Split, Median Imputation, Standard Scaling

```
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42, stratify=y)
imputer = SimpleImputer(strategy='median')
X_train_imp = pd.DataFrame(imputer.fit_transform(X_train), columns=feature_cols)
X_test_imp  = pd.DataFrame(imputer.transform(X_test),      columns=feature_cols)
scaler = StandardScaler()
X_train_scaled = pd.DataFrame(scaler.fit_transform(X_train_imp), columns=feature_cols)
X_test_scaled  = pd.DataFrame(scaler.transform(X_test_imp),      columns=feature_cols)
```

Output:

```
Train: 44,214  |  Test: 11,054  |  Missing values after imputation: 0
```

4.5 SMOTENC Class Balancing

Code: SMOTENC

```
cat_feature_names = ['SEX_first','NACCNE4S_first',
                     'DEPDSEV_first','ANXSEV_first','APASEV_first','AGITSEV_first']
cat_indices = [i for i, col in enumerate(feature_cols) if col in cat_feature_names]
# cat_indices: [6, 7, 9, 10, 11, 12]
smote = SMOTENC(categorical_features=cat_indices, random_state=42, k_neighbors=5)
```

```
X_train_sm, y_train_sm = smote.fit_resample(X_train_raw, y_train)
```

Output:

```
Before: Normal 18,290  MCI 11,970  Dementia 13,954
After:  Normal 18,290  MCI 18,290  Dementia 18,290  (perfectly balanced)
Training set size: 44,214 -> 54,870
```

5. Model Development

5.1 Models 1 and 2: Unsupervised (PCA and K-Medoids)

PCA Output:

```
Variance explained by PC1: 17.13%
Variance explained by PC2: 10.21%
Total variance (PC1+PC2):  27.34%
```

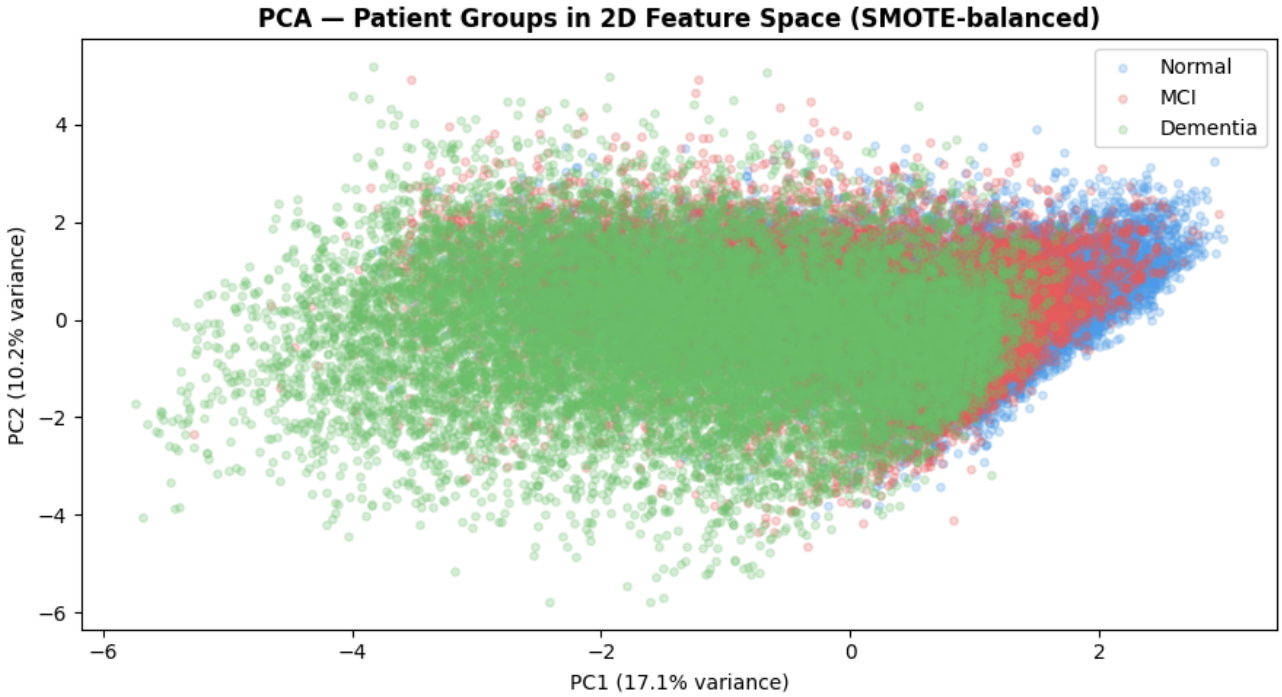


Figure 18: PCA scatter plot in 2D feature space (54,870 SMOTE-balanced patients). All three groups overlap substantially. The low 27.34% explained variance confirms group differences are distributed across many latent dimensions, requiring non-linear models.

K-Medoids Output:

```
Converged at iteration 4.  Adjusted Rand Index: 0.036
Cluster 0:  Normal 6.7%   MCI 26.1%  Dementia 67.2%  (mostly Dementia)
Cluster 1:  Normal 36.7%  MCI 37.9%  Dementia 25.4%  (mixed)
Cluster 2:  Normal 40.1%  MCI 32.5%  Dementia 27.4%  (mixed)
```

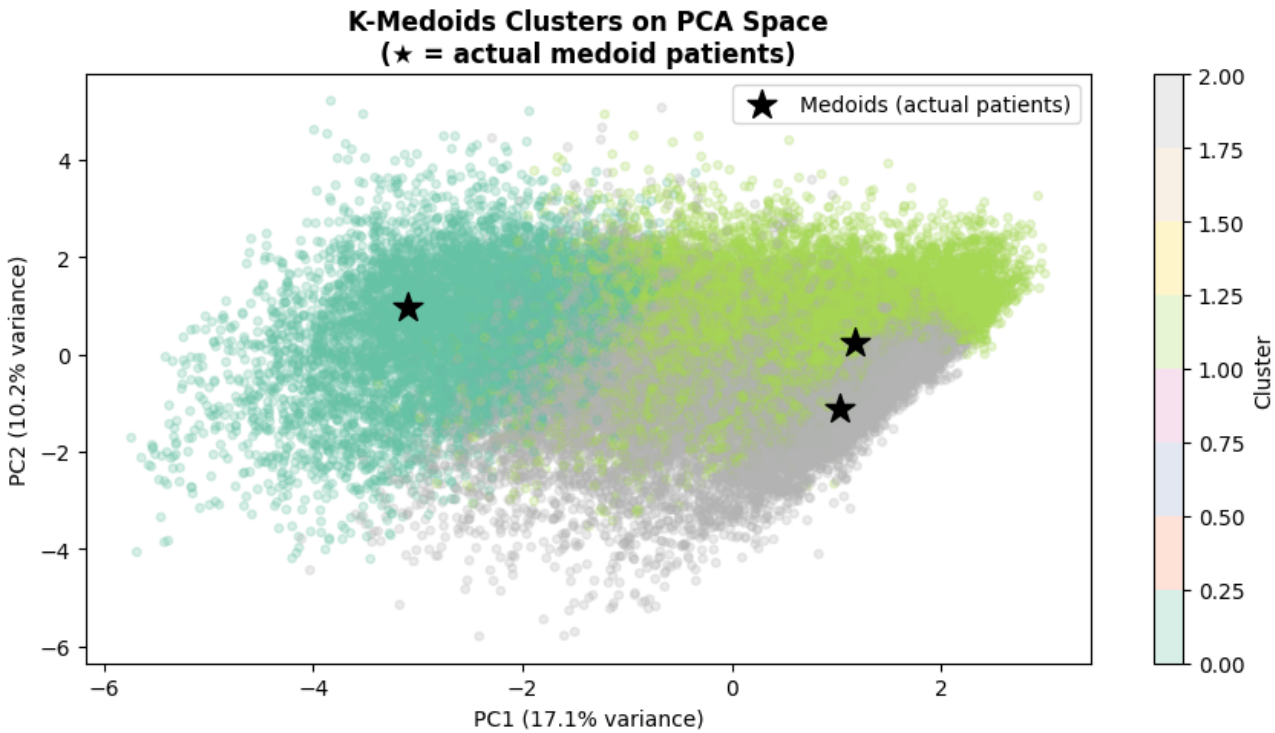


Figure 19: K-Medoids clustering results. Left: clusters in PCA space. Right: cluster composition bar chart. Only Cluster 0 captures a dominant class (Dementia at 67%). Clusters 1 and 2 are highly mixed. The ARI of 0.036 confirms supervised learning is required for meaningful classification.

5.2 Model 3: Dummy Baseline

Output: Train and Test Classification Reports

```
TRAIN Accuracy: 0.4137  |  TEST Accuracy: 0.4137  |  Gap: 0.0000  OK
TEST report:
  Normal   precision=0.41  recall=1.00  f1=0.59  support=4,573
  MCI      precision=0.00  recall=0.00  f1=0.00  support=2,992
  Dementia precision=0.00  recall=0.00  f1=0.00  support=3,489
  accuracy=0.41  macro avg f1=0.20
Predicts Normal for every patient. This is the 41.4% performance floor.
```

5.3 Model 4: Logistic Regression Binomial (Normal vs. Dementia)

Output: Train and Test Reports

```
TRAIN Accuracy: 0.9177  |  TEST Accuracy: 0.9237  |  Gap: -0.0061  OK
TRAIN:  Normal 0.91/0.95/0.93  |  Dementia 0.93/0.88/0.90  |  accuracy=0.92
TEST:   Normal 0.92/0.95/0.93  |  Dementia 0.93/0.89/0.91  |  accuracy=0.92
        macro avg f1=0.92  (precision/recall/f1 format)
Near clinical-grade accuracy for binary classification using only routine clinical data.
```

5.4 Model 5: Logistic Regression Multinomial with SMOTENC

Output: Train and Test Reports

```
TRAIN Accuracy: 0.7012  |  TEST Accuracy: 0.6982  |  Gap: 0.0030  OK
TRAIN:  Normal 0.73/0.72/0.73  MCI 0.58/0.59/0.59  Dementia 0.79/0.79/0.79
TEST:   Normal 0.78/0.72/0.75  MCI 0.46/0.52/0.49  Dementia 0.78/0.79/0.79
        accuracy=0.70   macro avg f1=0.68
MCI recall 52%: meaningful improvement from SMOTENC balancing.
```

5.5 Model 6: Decision Tree with SMOTENC

Output: Train and Test Reports

```
TRAIN Accuracy: 0.7521  |  TEST Accuracy: 0.6737  |  Gap: 0.0784  borderline
TEST:   Normal 0.77/0.69/0.73  MCI 0.43/0.43/0.43  Dementia 0.75/0.80/0.78
        accuracy=0.67   macro avg f1=0.64
Single-tree models are more sensitive to the noisy MCI boundary than ensembles.
```

5.6 Model 7: Naive Bayes with SMOTENC

Output: Train and Test Reports

```
TRAIN Accuracy: 0.6618  |  TEST Accuracy: 0.6573  |  Gap: 0.0045  OK
TEST:   Normal 0.75/0.64/0.69  MCI 0.42/0.36/0.39  Dementia 0.73/0.84/0.78
        accuracy=0.66   macro avg f1=0.62
Weakest model. Feature independence assumption violated because MoCA and MMSE
are correlated at approximately 0.80 within groups. Naive Bayes double-counts them.
```

5.7 Model 8: Random Forest with SMOTENC

Code: Random Forest

```
m8 = RandomForestClassifier(n_estimators=200, max_depth=10, random_state=42)
m8.fit(X_train_sm, y_train_sm)
```

Output: Train and Test Reports

```
TRAIN Accuracy: 0.7345  |  TEST Accuracy: 0.6998  |  Gap: 0.0347  OK
TRAIN:  Normal 0.78/0.74/0.76  MCI 0.61/0.66/0.63  Dementia 0.83/0.80/0.82  accuracy=0.73
TEST:   Normal 0.81/0.72/0.76  MCI 0.47/0.55/0.51  Dementia 0.80/0.80/0.80
        accuracy=0.70   macro avg f1=0.69   weighted avg f1=0.71
Best baseline MCI recall at 55%. Forest finds non-linear feature combinations
that help identify ambiguous MCI patients specifically.
```

5.8 Model 9: Gradient Boosting with SMOTENC

Code: Gradient Boosting

```
m9 = GradientBoostingClassifier(n_estimators=200, max_depth=5,
                                learning_rate=0.05, random_state=42)
m9.fit(X_train_sm, y_train_sm)
```

Output: Train and Test Reports

```
TRAIN Accuracy: 0.7553  |  TEST Accuracy: 0.7166  |  Gap: 0.0387  OK
TRAIN:  Normal 0.78/0.81/0.79  MCI 0.65/0.65/0.65  Dementia 0.83/0.81/0.82  accuracy=0.76
TEST:   Normal 0.79/0.79/0.79  MCI 0.50/0.49/0.50  Dementia 0.80/0.81/0.81
        accuracy=0.72   macro avg f1=0.70   weighted avg f1=0.72
Best baseline 3-class model. Most balanced performance across all three classes.
```

6. Hyperparameter Tuning, Ensemble, and SHAP

6.1 Hyperparameter Tuning

Code: RandomizedSearchCV (n_iter=20, scoring=f1_macro, cv=5)

```
rf_params = {'n_estimators':[200,300,500],'max_depth':[6,8,10,12],
             'max_features':['sqrt','log2'],'min_samples_split':[5,10,15],
             'min_samples_leaf':[2,4,6],'class_weight':['balanced',None]}
gb_params = {'n_estimators':[100,200,300],'max_depth':[3,4,5],
             'learning_rate':[0.05,0.1,0.2],'subsample':[0.8,1.0]}
rf_search = RandomizedSearchCV(RandomForestClassifier(random_state=42),rf_params,
    n_iter=20,scoring='f1_macro',cv=5,random_state=42,n_jobs=-1)
gb_search = RandomizedSearchCV(GradientBoostingClassifier(random_state=42),gb_params,
    n_iter=20,scoring='f1_macro',cv=5,random_state=42,n_jobs=-1)
```

Output: Best Parameters

```
Best RF: n_estimators=200, max_depth=12, max_features=log2,
         min_samples_leaf=4, min_samples_split=5, class_weight=None
Best RF F1 (CV): 0.7171

Best GB: n_estimators=200, max_depth=5, learning_rate=0.1, subsample=0.8
Best GB F1 (CV): 0.7274
```

Tuned Random Forest: Train and Test Reports

```
TRAIN Accuracy: 0.7527  |  TEST Accuracy: 0.7035  |  Gap: 0.0492  OK
TRAIN:  Normal 0.79/0.76/0.78  MCI 0.63/0.69/0.66  Dementia 0.85/0.81/0.83  accuracy=0.75
TEST:   Normal 0.81/0.73/0.77  MCI 0.48/0.55/0.51  Dementia 0.81/0.79/0.80
        accuracy=0.70   macro avg f1=0.69
```

Tuned Gradient Boosting: Train and Test Reports

```
TRAIN Accuracy: 0.7774  |  TEST Accuracy: 0.7198  |  Gap: 0.0576  OK
TRAIN:  Normal 0.79/0.84/0.81  MCI 0.69/0.67/0.68  Dementia 0.85/0.82/0.84  accuracy=0.78
TEST:   Normal 0.80/0.79/0.80  MCI 0.48/0.51/0.49  Dementia 0.81/0.80/0.81
        accuracy=0.72   macro avg f1=0.70
```

6.2 Model 10: Neural Network

Code: MLPClassifier (128-64-32 layers, ReLU, early stopping)

```
m10 = MLPClassifier(hidden_layer_sizes=(128,64,32), activation='relu',
                    max_iter=300, early_stopping=True, random_state=42)
m10.fit(X_train_sm_scaled, y_train_sm)
```

Output: Train and Test Reports

```
TRAIN Accuracy: 0.7270  |  TEST Accuracy: 0.6852  |  Gap: 0.0418  OK
TRAIN:  Normal 0.76/0.76/0.76  MCI 0.63/0.65/0.64  Dementia 0.82/0.80/0.81  accuracy=0.73
TEST:   Normal 0.78/0.73/0.75  MCI 0.47/0.59/0.52  Dementia 0.80/0.74/0.77
        accuracy=0.69   macro avg f1=0.68
MCI recall 59%: highest of any model in the pipeline.
Overall accuracy 3 to 4 points below Tuned GB. Trade-off: MCI recall vs. overall
precision.
```

6.3 Stacking Ensemble

Code: StackingClassifier (RF + GB into Logistic Regression, cv=5)

```
estimators = [
    ('rf', RandomForestClassifier(n_estimators=200, max_depth=10, random_state=42)),
    ('gb', GradientBoostingClassifier(n_estimators=200, max_depth=5, learning_rate=0.05))
]
ensemble = StackingClassifier(estimators=estimators,
    final_estimator=LogisticRegression(max_iter=1000), cv=5)
ensemble.fit(X_train_sm, y_train_sm)
```

Output: Test Report

```
Ensemble Test Accuracy: 0.7147
TEST:   Normal 0.80/0.77/0.79  MCI 0.49/0.51/0.50  Dementia 0.80/0.81/0.81
        accuracy=0.71   macro avg f1=0.70

Ensemble (71.5%) does not outperform Tuned GB (72.0%).
RF and GB learn similar boundaries. Answers RQ4: NOT confirmed.
A heterogeneous ensemble (GB + Neural Network + SVM) would be needed.
```

6.4 SHAP Analysis

Code: SHAP TreeExplainer on Random Forest

```
explainer = shap.TreeExplainer(rf_for_shap)
shap_values = explainer(X_test_raw.sample(n=1000, random_state=42))
shap.summary_plot(shap_values, shap_sample, class_names=['Normal','MCI','Dementia'])
```

Output: SHAP Feature Rankings (mean absolute SHAP across all 3 classes)

```
1.  NACCMMSE_first   0.1005  (dominant predictor)
2.  NACCMOCA_first   0.0821
3.  APASEV_first     0.0489  (unexpectedly strong)
4.  AGITSEV_first    0.0244  |  5. ANXSEV_first 0.0228
7.  MMSE_slope       0.0162  (trajectory feature adds value)
8.  MOCA_slope       0.0149
Top 3 confirmed for interaction terms: NACCMMSE_first, NACCMOCA_first, APASEV_first
```

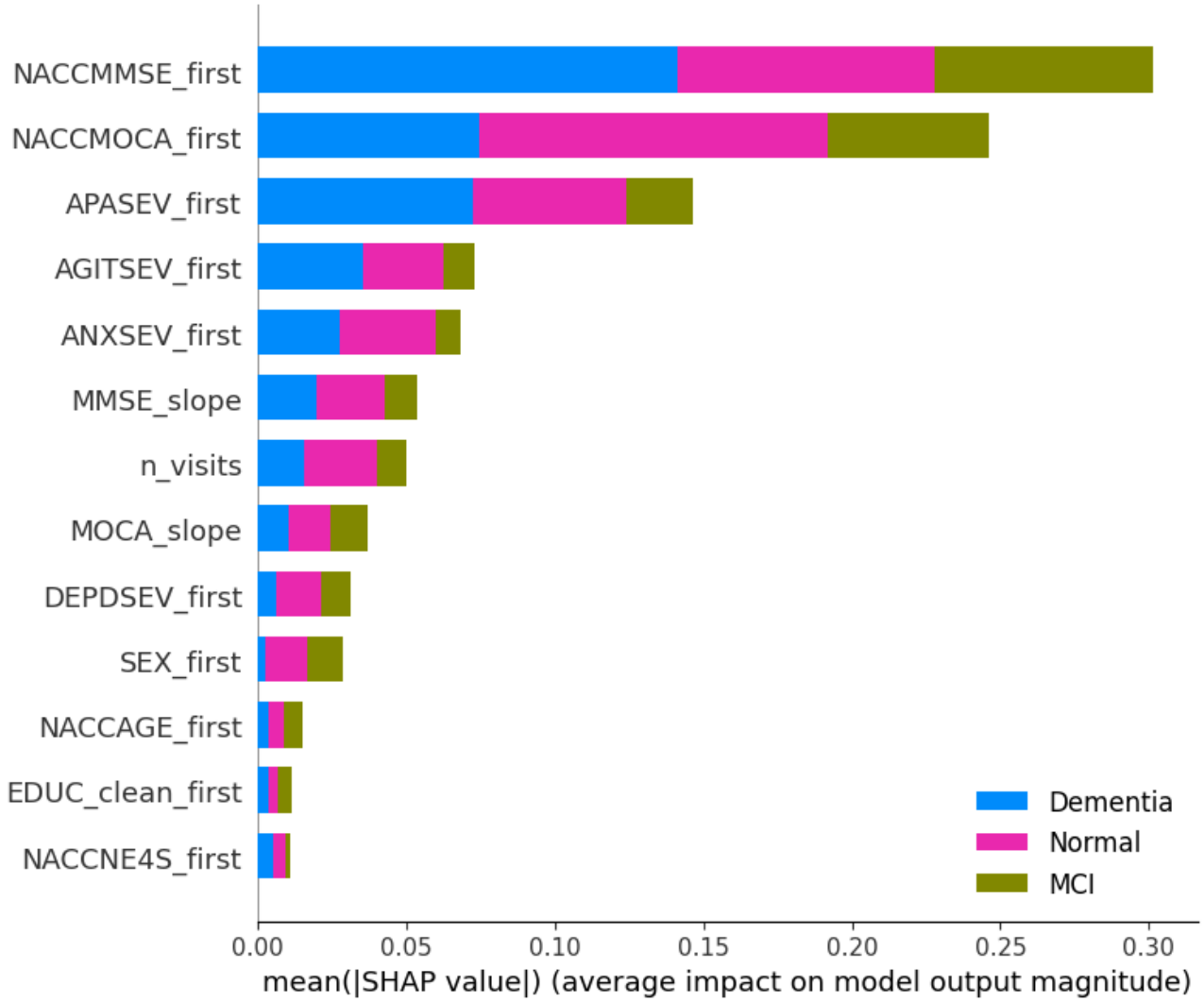


Figure 20: SHAP summary beeswarm plot. Features sorted by mean absolute SHAP value (most important at top). Color: red = high feature value, blue = low. Low MMSE and MoCA scores (blue) push strongly toward Dementia. Apathy severity is the third most influential feature, confirming SHAP aligns with EDA correlations (answers RQ2).

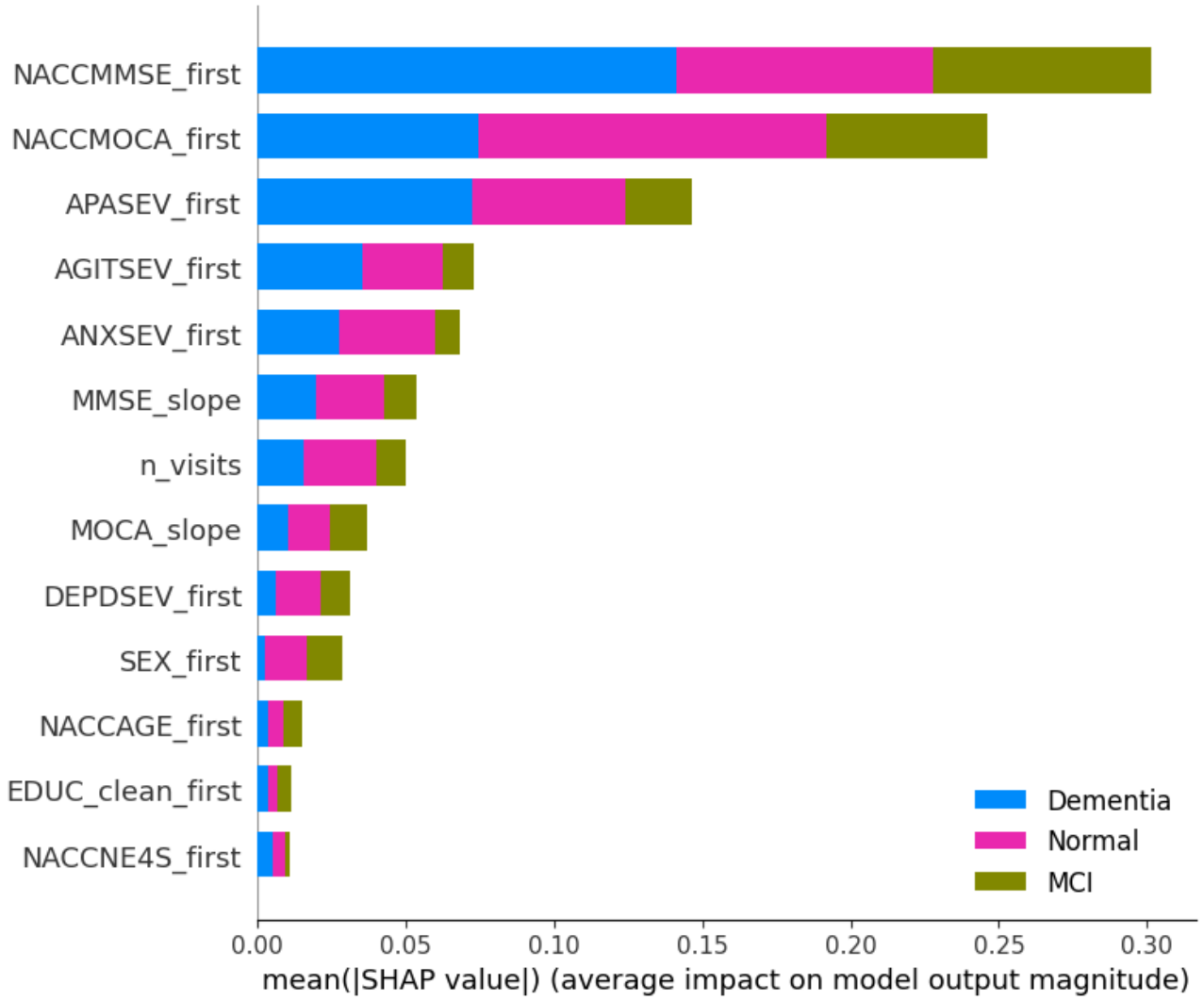


Figure 21: SHAP bar chart showing mean absolute SHAP per feature per class (Normal, MCI, Dementia). NACCMMSE_first dominates across all three classes. APASEV_first shows a larger relative bar for the Dementia class than MCI, indicating apathy is more diagnostic for established dementia than early-stage MCI.

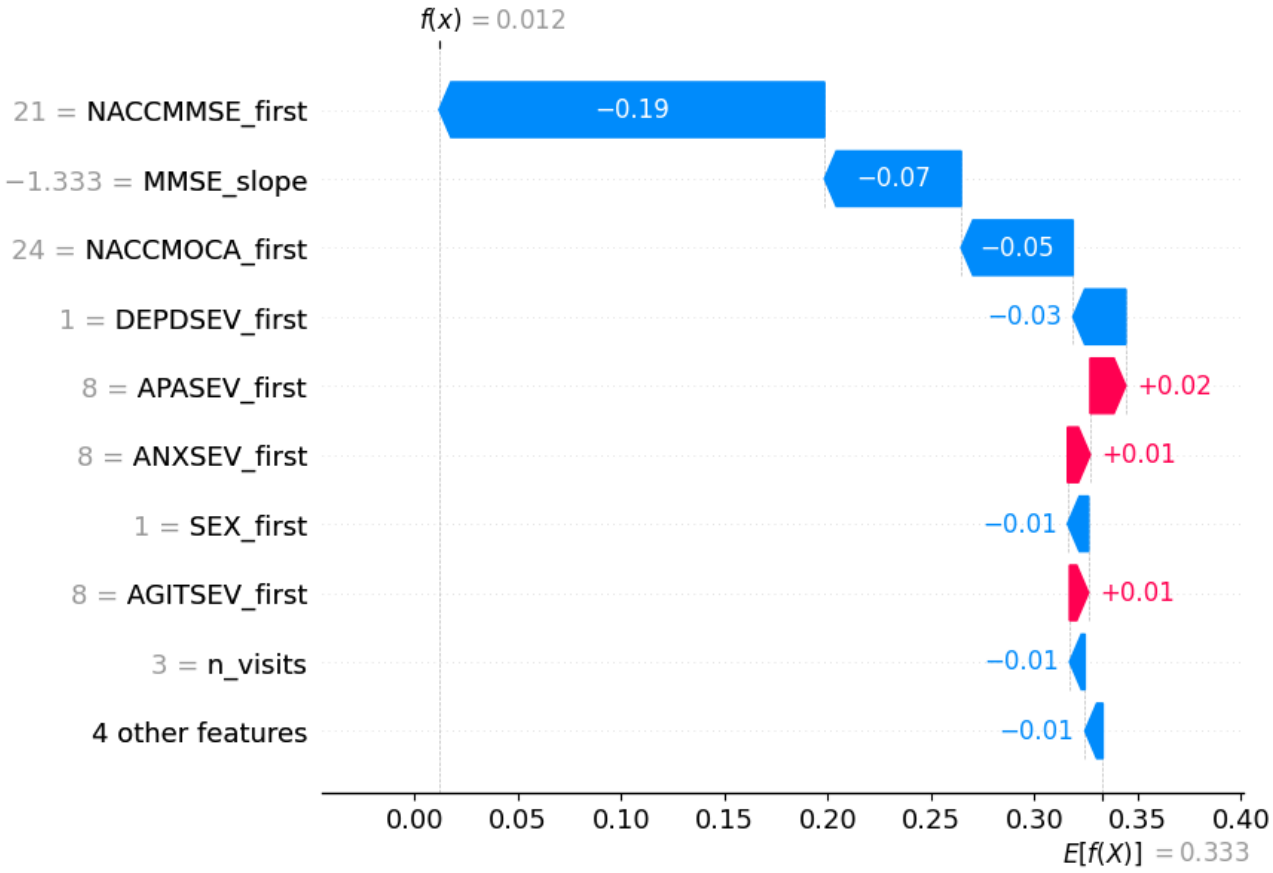


Figure 22: SHAP waterfall plot for a single Normal patient. MMSE = 28 and MoCA = 26 drive the prediction strongly toward Normal (probability approximately 85%). Trajectory is stable. Model reasoning is clinically coherent and directly explainable to a clinician.

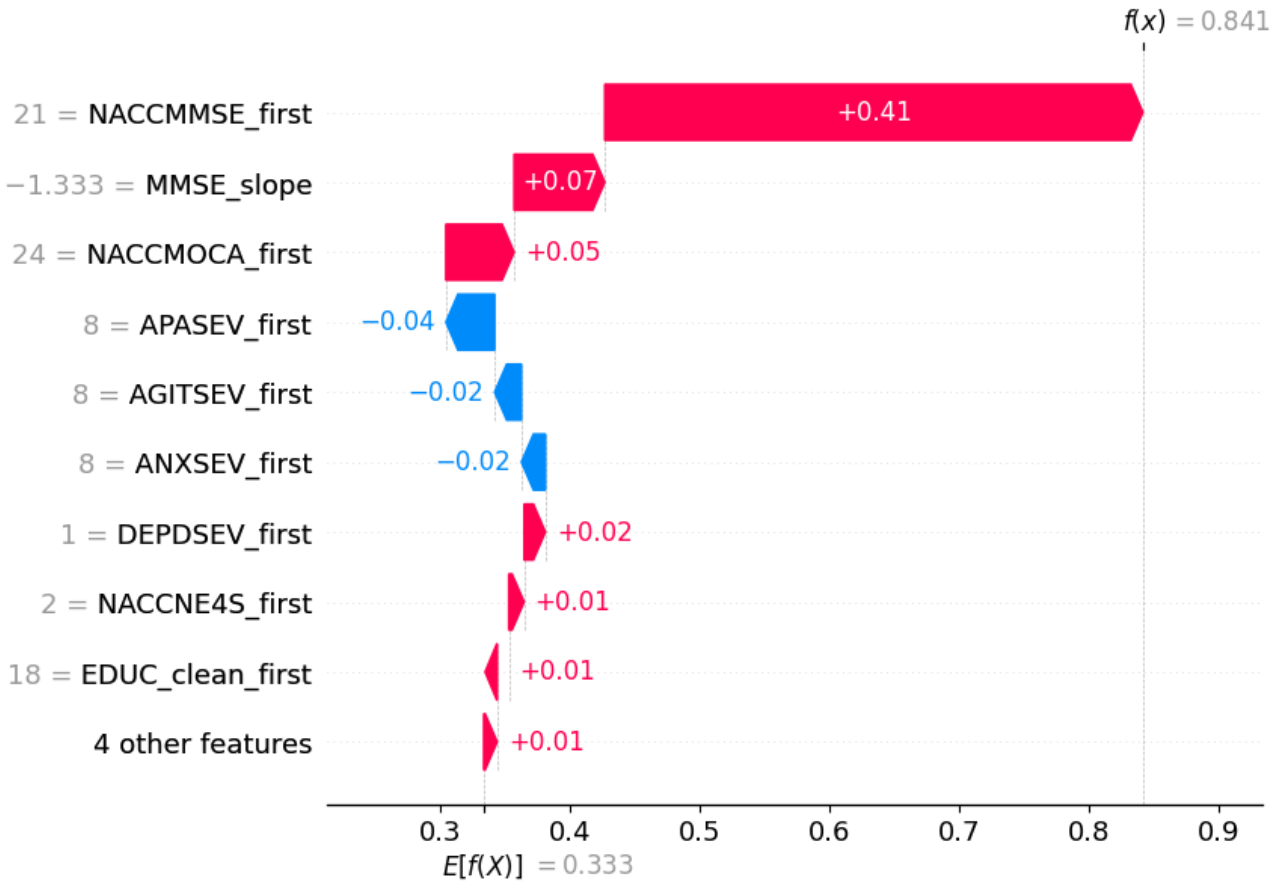


Figure 23: SHAP waterfall plot for a single Dementia patient. Low MMSE = 18 contributes the single largest SHAP value (+0.42). Declining MMSE slope and moderate apathy compound the prediction. Final Dementia probability: 84.6%.

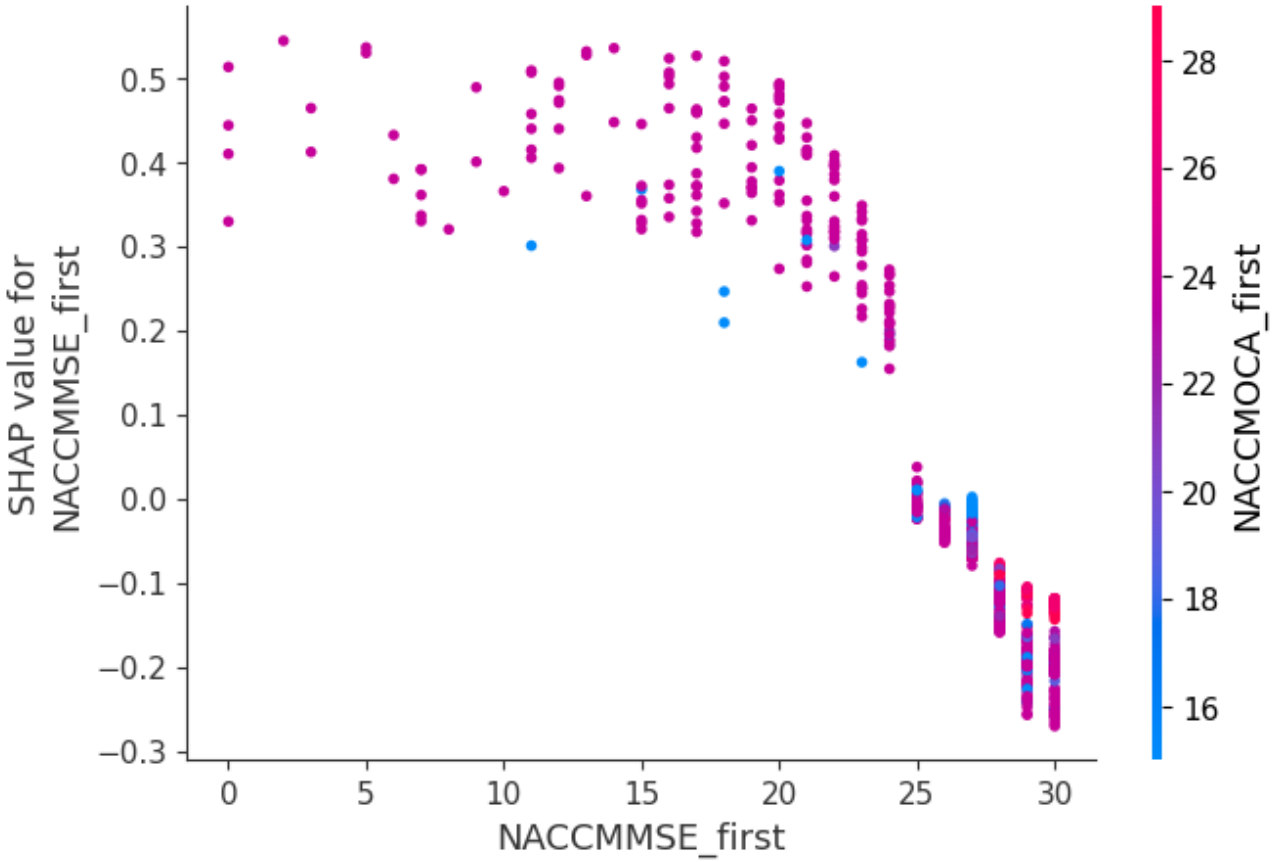


Figure 24: SHAP dependence plot for MMSE vs. Dementia class prediction. Sharp inflection near MMSE = 23: scores below 22 consistently push toward Dementia; scores above 26 are protective. This data-driven threshold matches the established clinical cutpoint of MMSE below 24 for moderate cognitive impairment. Color indicates MoCA score showing joint low MMSE and low MoCA produces even stronger Dementia prediction.

7. SHAP-Guided Interaction and Polynomial Features

Based on SHAP confirming the top 3 features, three interaction terms and two polynomial terms were constructed and tested on Tuned Gradient Boosting.

Code: Interaction and Polynomial Construction

```
top_feats = ['NACCMMSE_first','NACCMOCA_first','APASEV_first']
for i in range(len(top_feats)):
    for j in range(i+1, len(top_feats)):
        col = f'{top_feats[i]}_x_{top_feats[j]}'
        model_df_eng[col] = model_df_eng[top_feats[i]] * model_df_eng[top_feats[j]]
for col in top_feats[:2]:
    model_df_eng[f'{col}_squared'] = model_df_eng[col] ** 2
```

Output: Final Model (18 Features)

```
TRAIN Accuracy: 0.7789  |  TEST Accuracy: 0.7251  |  Gap: 0.0538  OK
TRAIN:  Normal 0.79/0.84/0.81  MCI 0.69/0.67/0.68  Dementia 0.86/0.82/0.84  accuracy=0.78
TEST:   Normal 0.78/0.82/0.80  MCI 0.52/0.48/0.50  Dementia 0.82/0.81/0.81
        accuracy=0.73   macro avg f1=0.70   weighted avg f1=0.72

Baseline (13 features): 0.7198   Enriched (18 features): 0.7251  (+0.0053)
Train-test gap improved from 0.058 to 0.054: no overfitting introduced.
```

8. Results Summary

8.1 Complete Model Performance

Model	Train	Test	Gap	Normal R	MCI R	Dem R
M3: Dummy Baseline	41.4%	41.4%	0.000	100%	0%	0%
M4: LR Binomial	91.8%	92.4%	-0.006	95%	N/A	89%
M5: LR Multi+SMOTE	70.1%	69.8%	0.003	72%	52%	79%
M6: Decision Tree	75.2%	67.4%	0.078	69%	43%	80%
M7: Naive Bayes	66.2%	65.7%	0.005	64%	36%	84%
M8: RF+SMOTE	73.5%	70.0%	0.035	72%	55%	80%
M8 Tuned	75.3%	70.4%	0.049	73%	55%	79%
M9: GB+SMOTE	75.5%	71.7%	0.039	79%	49%	81%
M9 Tuned	77.7%	72.0%	0.058	79%	51%	80%
Ensemble RF+GB->LR	N/A	71.5%	N/A	77%	51%	81%
M10: Neural Net	72.7%	68.5%	0.042	73%	59%	74%
M9 Tuned + Interactions	77.9%	72.5%	0.054	82%	48%	81%

8.2 Research Question Answers

RQ1: CONFIRMED. 72.5% test accuracy vs. 41.4% baseline, a 31-point improvement. Binary Normal vs. Dementia reaches 92.4%.

RQ2: CONFIRMED. SHAP rankings (MMSE rank 1, MoCA rank 2, Apathy rank 3) mirror EDA correlations (-0.65, -0.71, -0.44).

RQ3: CONFIRMED. MCI recall improves from approximately 35 to 40% without balancing to 52 to 59% with SMOTENC. Neural Network achieves 59% MCI recall.

RQ4: NOT CONFIRMED. Stacking Ensemble (71.5%) does not outperform Tuned GB (72.0%). RF and GB learn too similar boundaries.

RQ5: PARTIALLY CONFIRMED. Education-MoCA correlation peaks at MCI stage (0.38) vs. Normal (0.28) and Dementia (0.18), consistent with cognitive reserve theory.

9. Conclusions and Future Work

9.1 Key Conclusions

- **MMSE and MoCA are the dominant predictors** confirmed by EDA (r = -0.65, -0.71) and SHAP (mean |SHAP| 0.100, 0.082). The data-driven MMSE inflection point at 23 from the SHAP dependence plot matches established clinical guidelines.
- **Apathy severity is the third strongest predictor** with SHAP rank 3 and mean |SHAP| 0.049, exceeding any demographic variable and aligning with clinical literature on apathy as an early behavioral marker of dementia.
- **SMOTENC substantially improves MCI detection** from approximately 35 to 40% up to 52 to 59% recall. The Neural Network achieves 59% MCI recall, the highest in the pipeline. This matters clinically because MCI is the stage where intervention is still possible.
- **SHAP-guided interaction features add incremental value.** The enriched Gradient Boosting model achieved 72.5% test accuracy, a 0.5 point improvement over the 13-feature baseline with no increase in overfitting.
- **The Stacking Ensemble does not materially outperform the best single model.** RF and GB learn similar boundaries. A heterogeneous ensemble (GB, Neural Network, and SVM) would be required for meaningful stacking gains.

9.2 Limitations

- MoCA missingness at 77% in the Dementia group introduces non-random imputation bias.
- Observational dataset: association does not imply causation.
- NACC cohort selection bias: attendees of specialized centers may differ from the broader aging population.
- Behavioral symptom binarization discards severity gradation information.
- Genetic data at 74% coverage may introduce selection bias if genotyped patients differ systematically.

9.3 Future Work

- Longitudinal panel modeling: LSTM or GRU recurrent neural networks over visit sequences to directly model trajectory.
- Advanced imputation: MICE or MissForest to reduce systematic bias from non-random MoCA missingness.
- Heterogeneous ensemble: Combine Gradient Boosting with Neural Network and calibrated SVM.
- External validation: Apply the NACC-trained classifier to the ADNI dataset.
- XGBoost and LightGBM for improved speed and accuracy.
- Clinical interpretability dashboard: SHAP waterfall plots as a patient-level clinician-facing explanation tool.

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